Q-Learning with Function Approximation

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As before, given a behavioral policy π, we obtain a dataset

$$D = \{ (s_0, a_0, r_0), (s_1, a_1, r_1), \ldots\ldots \}$$

by applying π to the system.

Now, we will assume that

$$Q(s,a) = f_\theta(s,a), \quad \text{where } f_\theta \text{ is}$$

a function parametrized by $\theta \in \mathbb{R}^d$, where $d << |S||A|$

For example, f_θ could be a neural network or it could be $\phi^T(s,a)\theta$ where $\phi(s,a)$ is a known feature vector. Suppose we have an estimate θ_k of θ, how do we get a new estimate θ_{k+1}. A natural algorithm would be to choose θ_{k+1} to solve

$$\min_\theta \frac{1}{2} \| f_\theta - T(f_{\theta_k}) \|^2_{d_\pi}$$

where Φ is matrix whose rows at $\phi^T(s,a)$

and $\|Q\|^2_{d_\pi} = \sum_{s,a} d_\pi(s,a) Q^2(s,a)$ and $d_\pi(s,a)$ is the stationary probability of (s,a)

$$\Leftrightarrow \min_\theta \underbrace{\frac{1}{2} \sum_{s,a} d_\pi(s,a) \Big(f_\theta(s,a) - r(s,a) - \alpha E\big(\max_{a'} f_{\theta_k}(s,a')\big)\Big)^2}_{\triangleq J_\theta}$$

$$\nabla_\theta J_\theta = \sum_{s,a} d_\pi(s,a) \Big(f_\theta(s,a) - r(s,a) - \alpha E\big(\max_{a'} f_{\theta_k}(s,a')\big)\Big) \nabla_\theta f_\theta(s,a)$$

$$= \sum_{s,a,s'} d_\pi(s,a) P(s'|s,a) \Big(f_\theta(s,a) - r(s,a) - \alpha \max_{a'} f_{\theta_k}(s,a')\Big) \nabla_\theta f_\theta(s,a)$$

If we pick a (s_t, a_t, s_{t+1}) from the dataset, then approximately

$$(s_t, a_t, s_{t+1}) \sim d_\pi(s_t, a_t) P(s_{t+1} | s_t, a_t)$$

Thus, by uniform sampling,

$$\nabla_\theta J_\theta = \frac{1}{T-1} \sum_{t=0}^{T-1} \Big(f_\theta(s_t,a_t) - r(s_t,a_t) - \alpha \max_{a'} f_{\theta_k}(s_t,a')\Big) \nabla_\theta f_\theta(s_t,a_t)$$

Thus, SGD to obtain θ_{k+1} would be

$$\boxed{\hat{\theta}_{k+1,t+1} = \hat{\theta}_{k+1,t} - \varepsilon_t \Big(f_\theta(s_t,a_t) - r(s_t,a_t) - \alpha \max_{a'} f_{\theta_k}(s_t,a')\Big) \nabla_\theta f_\theta(s_t,a_t), \quad \hat{\theta}_{k+1,0} \text{ arbitrary}}$$

SGD

So, the overall algorithm is as follows:

- Choose θ_0, set $k=0$

- Execute SGD for $t = \{0, 1, \ldots, T-1\}$
- $\theta_{k+1} = \hat{\theta}_{k+1,T}$

- Set $k \leftarrow k+1$

□

Outline of why the algorithm works:

- Suppose SGD gives the correct θ_{k+1}. This is hard to prove in general. But if $f_\theta(s,a) = \phi^T(s,a)\theta$, then we have SGD for a quadratic loss function, and one can show convergence.

- Assuming SGD converges, $f(\theta_{k+1})$ can be thought of as the best approximation to Q_{k+1} given by

$$Q_{k+1} = T(f_{\theta_k})$$

In other words,

$$f_{\theta_{k+1}} = T(f_{\theta_k}) + e_k \quad \text{where } e_k \text{ is the error.}$$

$$\Rightarrow \|f_{\theta_{k+1}} - Q^*\|_\infty = \|T(f_k) + e_k - Q^*\|_\infty$$

$$\leq \|T(f_k) - Q^*\|_\infty + \|e_k\|_\infty$$

$$\leq \alpha \|f_k - Q^*\|_\infty + e, \quad \text{assume } \|e_k\|_\infty \leq e \ \forall k$$

$$\Rightarrow \boxed{\lim_{k \to \infty} \|f_k - Q^*\|_\infty \leq \frac{e}{1-\alpha}}$$

So if e is small, we have a good approx. to Q^*.

These heuristic arguments can be made precise for the case $f_\theta = \Phi\theta$ and with other minor changes to the algorithm.

See Chen, Clarke, Maguluri (2023, SIAM J. Data Science)